Skeleton-Aware Graph Embeddings for Multi-Person Pose Grouping

by

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SUMMARY

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LIST OF ABBREVIATIONS

ANN	artificial neural network
EMI	electromagnetic interference
MCMC	Markov chain Monte Carlo
SVM	support vector machine
Note:	Make sure the list of abbreviations is sorted alphabetically. This list is formatted as a table with no borders. Insert rows as needed.

LIST OF SYMBOLS [OPTIONAL]

I	current
R	resistance
V	voltage
Note:	Make sure the list of symbols is sorted alphabetically, and/or according to the Greek alphabet. The list is formatted as a table with no borders. Insert rows as needed.

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CHAPTER 1 INTRODUCTION

This chapter motivates multi-person pose grouping as a separable problem within bottom-up human pose estimation (HPE), identifies the gap in modular detector-agnostic grouping methods, and sets out the research questions, approach, and contributions that the rest of the dissertation defends.

1.1 CONTEXT

Introduces the detection/grouping split in bottom-up HPE: modern detectors (HigherHRNet, HRNet) produce accurate per-joint heatmaps; assigning those joints to person identities is the remaining bottleneck. Summarises why joint-to-person assignment is intrinsically difficult (unknown K, type constraints, occlusion, scale) and why it deserves to be studied as a module in its own right.

1.2 PROBLEM STATEMENT

Formulates the task: given N detected keypoints with joint-type labels $t_i \in \{1, \ldots, 17\}$ from an image containing an unknown number of people K, assign each keypoint to one of K person groups under the constraint that every person has at most one keypoint of each type.

1.3 RESEARCH GAP

Existing grouping methods (associative embeddings in HigherHRNet, part-affinity fields in OpenPose, HGG, CenterGroup) are tightly coupled to their host detectors and trained jointly with them. A *modular* grouping module that operates on any detector's keypoint output is largely unstudied. This is the gap this work addresses.

1.4 RESEARCH OBJECTIVE AND RESEARCH QUESTIONS

The objective is to design and evaluate a detector-independent, GNN-based pose grouping module. The investigation addresses the following research questions. Bracketed short answers anticipate the findings from Chapter 4; full support is provided there.

- **RQ1 — The lever.** Is the bottleneck for clustering-based pose grouping the embedding, or the grouping algorithm? *[Answer: the embedding. Four learned grouping heads (DEC, slot attention, graph partitioning, SA-DMoN) fail to beat plain kNN on the same embeddings; strengthening the embedding (PG-GAT) then beats every prior configuration.]*
- **RQ2 — Inductive biases.** Which skeleton-aware modifications to a GATv2 embedding improve pose grouping, and by how much? *[Answer: three modifications — type-pair attention modulation, skeleton-aware positional encoding, and a type-repulsion loss term — each contribute independently; the full combination is the best configuration.]*
- **RQ3 — Detector independence.** Can a grouping module trained separately from the detector match an integrated end-to-end grouping mechanism (HigherHRNet's associative embedding head) on a real-image benchmark? *[Answer: not fully. PG-GAT + kNN end-to-end is approximately 0.04 PGA below HigherHRNet AE on COCO, with detector-independence as the trade-off.]*
- **RQ4 — Embedding geometry.** Does the choice of embedding space (Euclidean vs hyperbolic) affect pose grouping quality? *[Answer: yes before COCO fine-tuning, where hyperbolic at* $d = 32$ *matches Euclidean at* $d = 128$ *(an efficiency edge). After fine-tuning the advantage disappears — reported as an efficiency trade, not a quality win.]*
- **RQ5 — Graph primitive.** Is a higher-order graph primitive (skeleton triplets) a better node for pose grouping than single joints? *[Answer: no. TriGAT underperforms joint-level PG-GAT on every comparison, with the voting step (triplet clusters to joint labels) as the dominant failure mode.]*
- **RQ6 — Fine-tuning.** How much does COCO fine-tuning lift a synthetically-trained embedding, and does it damage performance on the synthetic domain? *[Answer: +7.5 PGA on COCO (kNN), the single largest gain of any experiment in this work; synthetic performance is not damaged (0.910* $\rightarrow$ *0.928).]*

1.5 APPROACH

Train a graph attention network with contrastive loss on detected keypoints; cluster at test time with kNN. The iteration order mirrors the investigation: standard GAT baseline $\rightarrow$ a series of learned grouping heads (DEC, slot attention, graph partitioning), all of which failed to beat kNN $\rightarrow$ SA-DMoN, a skeleton-aware modularity objective, which also failed $\rightarrow$ PG-GAT, which keeps the clustering head simple but strengthens the embedding with three skeleton-aware modifications. Evaluate on synthetic data (controlled K, perfect GT) and COCO val2017 (zero-shot and fine-tuned). Close the loop by integrating with HigherHRNet for end-to-end validation.

1.6 RESEARCH CONTRIBUTIONS

The contributions of this work are:

1. **PG-GAT**, a skeleton-aware graph attention network for pose-keypoint embedding. Three modifications over standard GATv2: type-pair attention modulation, skeleton-aware positional encoding, and a type-repulsion term in the contrastive loss.
2. **Empirical evidence that the embedding, not the grouping head, is the decisive lever** for clustering-based pose grouping. Documented across four learned-head approaches (DEC, slot attention, graph partitioning, and ten SA-DMoN variants), none of which outperformed plain kNN on the same embeddings.
3. **A COCO fine-tuning protocol** that lifts PG-GAT to 0.957 PGA on COCO val2017 with kNN clustering, and end-to-end to 0.942 when paired with HigherHRNet detection.
4. **An alternative-embedding-space study** (hyperbolic PG-GAT in the Lorentz model): hyperbolic at $d = 32$ matches Euclidean at $d = 128$ pre-fine-tune, but the advantage disappears after COCO fine-tuning — documented as an efficiency trade, not a quality win.
5. **An alternative-primitive study** (TriGAT, triplet nodes): a negative result reported with the failure-mode analysis (voting step propagates errors).

1.7 RESEARCH OUTPUTS

The primary research output is a journal paper on PG-GAT, targeted at *Image and Vision Computing* (Q1, IF 5.34), with *Pattern Recognition* (Q1, IF 9.84) as a stretch target.

1.8 OVERVIEW OF STUDY

Chapter 2 reviews the literature on bottom-up human pose estimation, graph neural networks, and clustering methods relevant to pose grouping. Chapter 3 presents the mathematical preliminaries (GNN, GCN, GAT/GATv2, DEC, slot attention, graph partitioning, DMoN) and describes the methodology behind each method investigated, culminating in PG-GAT. Chapter 4 reports experimental results across the methods. Chapter 5 discusses what the results reveal about the problem. Chapter 6 concludes and indicates future work.

1.9 DECLARATION REGARDING USE OF GENERATIVE AI

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CHAPTER 2 LITERATURE REVIEW

This chapter situates multi-person pose grouping within the broader human pose estimation literature, reviews the existing families of grouping mechanisms, and surveys the graph neural network and representation-learning work that PG-GAT builds on. The review closes by identifying the specific gap that this dissertation addresses.

2.1 MULTI-PERSON HUMAN POSE ESTIMATION

Overview of top-down (detect-person-then-pose) versus bottom-up (detect-joints-then-group) pipelines. Why bottom-up is attractive (constant time in number of people, robust to overlapping detections) and why it reintroduces the grouping problem. Canonical detectors: Hourglass networks, HRNet, HigherHRNet.

2.2 GROUPING IN BOTTOM-UP POSE ESTIMATION

Review of the major grouping mechanisms in the bottom-up literature: part-affinity fields (OpenPose), associative embeddings (Newell et al.), PersonLab mid-range offsets, HGG (graph-based), CenterGroup. How each binds tightly to its host detector and why that limits modularity.

2.3 GRAPH NEURAL NETWORKS FOR POSE

Existing graph-based approaches in the pose literature: STGAT (skeleton-based action recognition), GAST-Net (single-person 3D pose from video), skeleton-aware GCN variants. How these differ from the multi-person keypoint grouping task — none of the existing “skeleton-aware” networks target grouping, which motivates both the PG-GAT contribution and the naming choice (avoiding collision with STGAT/GAST-Net descriptors).

2.4 REPRESENTATION LEARNING FOR CLUSTERING

Contrastive learning (InfoNCE, SimCLR), triplet losses, metric learning. Deep clustering lineage: DEC (Xie et al.), IDEC, DeepCluster, DMoN. What these methods assume about the feature space and how those assumptions interact with the pose-grouping task.

2.5 GEOMETRIC EMBEDDING SPACES

Euclidean versus non-Euclidean embedding spaces. Hyperbolic embeddings (Poincaré ball, Lorentz model) and their suitability for tree-structured data. Relevance to skeletons, which are trees. Sets up the motivation for the hyperbolic PG-GAT variant evaluated in Chapter 4.

2.6 RESEARCH GAP

Synthesises the review: modular grouping modules for bottom-up HPE are under-explored; existing graph-based pose networks target different tasks; the clustering-head literature has not been systematically applied to pose grouping; and the choice of embedding space (Euclidean vs hyperbolic) is untested in this domain. PG-GAT and its ablations address these gaps.

CHAPTER 3 METHODOLOGY

This chapter presents the mathematical preliminaries that every method in this dissertation builds on, the problem formulation and data pipeline, and the methodology for each approach investigated — from the standard GAT baseline, through the series of learned grouping heads that were tried and failed, through SA-DMoN, and finally to PG-GAT. Two variants (hyperbolic PG-GAT, TriGAT) and the integration with a realistic detector are described last.

3.1 PRELIMINARIES — GRAPH NEURAL NETWORKS

A human skeleton is a graph: seventeen keypoints form the nodes and the skeletal connections form the edges. A graph neural network (GNN) embeds each node while conditioning on the structure of the graph — the natural choice for producing pose-grouping embeddings that are aware of joint-type and connectivity information. This section introduces the message-passing framework that unifies modern GNN variants, derives the graph convolutional network (GCN) as the canonical spectral instance, and develops the graph attention network (GAT), culminating in the GATv2 correction that is used as the attention primitive in every experiment in this dissertation.

3.1.1 The Message-Passing Framework

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote an undirected graph with N nodes and edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. Each node $i \in \mathcal{V}$ is associated with a feature vector $\mathbf{h}_i^{(0)} \in \mathbb{R}^{d_0}$, and each edge $(i, j) \in \mathcal{E}$ optionally carries an edge feature $\mathbf{e}_{ij}$. The neighbourhood of node i is $\mathcal{N}(i) = \{j : (i, j) \in \mathcal{E}\}$.

Almost every modern GNN variant can be expressed as iterated message passing [1]. At layer $l+1$, each node receives a message from each of its neighbours, aggregates them with a permutation-invariant operator, and updates its representation accordingly. Formally,

$$\mathbf{m}_{i \leftarrow j}^{(l+1)} = \mathrm{MSG}^{(l)}\left(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{e}_{ij}\right), \tag{3.1}$$

$$\mathbf{h}_i^{(l+1)} = \mathrm{UPDATE}^{(l)}\left(\mathbf{h}_i^{(l)}, \bigoplus_{j\in\mathcal{N}(i)} \mathbf{m}_{i\leftarrow j}^{(l+1)}\right), \tag{3.2}$$

where $\bigoplus$ denotes a permutation-invariant aggregator (sum, mean, or max) and $\mathrm{MSG}^{(l)}$ and $\mathrm{UPDATE}^{(l)}$ are learnable functions at layer l. Different choices of these three components recover different GNN architectures. GCN (Section 3.1.2) uses a linear message, a degree-based weighted sum aggregator, and a linear–nonlinear update. GAT (Section 3.1.3) introduces learned attention weights on the aggregator, making each neighbour's contribution data-dependent. GraphSAGE [2] uses concatenated mean–max aggregation with a feedforward update, optimised for inductive settings.

The message-passing view clarifies two points that matter for the remainder of this chapter. First, the inductive biases of a GNN are encoded in the choice of aggregator and the edge-weighting scheme, not in the overall template. Second, the number of layers controls the receptive field — a node's representation after L layers aggregates information from its L-hop neighbourhood — which, for skeleton graphs with at most four hops between any two joints, places a natural upper bound on depth.

3.1.2 Graph Convolutional Networks (GCN)

The GCN architecture of [3] derives from a first-order approximation of spectral graph convolutions. Given adjacency matrix $A \in \mathbb{R}^{N\times N}$ and degree matrix $D = \mathrm{diag}(\sum_j A_{ij})$, the symmetric normalised adjacency with self-loops is

$$\tilde{A} = \tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}, \tag{3.3}$$

where $\tilde{D} = \mathrm{diag}(\sum_j (A+I)_{ij})$ is the degree matrix of the self-loop-augmented graph. The GCN propagation rule is

$$H^{(l+1)} = \sigma\left(\tilde{A}H^{(l)}W^{(l)}\right), \tag{3.4}$$

where $H^{(l)} \in \mathbb{R}^{N\times d_l}$ stacks the node features row-wise, $W^{(l)} \in \mathbb{R}^{d_l\times d_{l+1}}$ is a learnable weight matrix, and $\sigma(\cdot)$ is a pointwise nonlinearity.

Two properties of Equation (3.4) motivate the move to attention-based aggregation in the pose-grouping setting. First, the edge weights are fixed: once $\tilde{A}$ is constructed from the graph topology, no learning signal modulates the contribution of individual neighbours. Two neighbours of the same target node contribute in proportion determined only by the degrees of the three nodes involved. Second, this uniform treatment is informationally inadequate for skeleton graphs where edge semantics vary sharply: the edge from the left shoulder to the left elbow encodes different information from the edge from the left shoulder to the head, and a pose-grouping embedding should be free to weight them differently. Attention-based aggregation, introduced next, provides exactly this flexibility.

3.1.3 Graph Attention Networks (GAT and GATv2)

GAT [4] replaces the fixed normalisation of Equation (3.3) with a learned attention weight for each edge. For a pair (i, j) with $j \in \mathcal{N}(i)$, the unnormalised attention score is

$$e_{ij}^{\text{GAT}} = \text{LeakyReLU}\Big(\mathbf{a}^\top \left[W\mathbf{h}_i \,\|\, W\mathbf{h}_j\right]\Big), \tag{3.5}$$

where $W \in \mathbb{R}^{d' \times d}$ is a shared linear projection, $\mathbf{a} \in \mathbb{R}^{2d'}$ is a learned attention vector, and $[\cdot\|\cdot]$ denotes vector concatenation. The scores are normalised over the neighbourhood with a softmax,

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}, \tag{3.6}$$

and the updated node representation is the attention-weighted sum of projected neighbour features,

$$\mathbf{h}_i' = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W\mathbf{h}_j\right). \tag{3.7}$$

Multi-head attention is recovered by repeating Equations (3.5)–(3.7) with independent parameters for each head and concatenating or averaging the outputs.

Equation (3.5) exposes a subtle limitation that was identified by [5] and has direct consequences for pose grouping. Because $\mathbf{a}^\top$ is applied to the concatenation $[W\mathbf{h}_i \| W\mathbf{h}_j]$, the attention score decomposes as $\mathbf{a}_1^\top W\mathbf{h}_i + \mathbf{a}_2^\top W\mathbf{h}_j$ where $\mathbf{a} = [\mathbf{a}_1; \mathbf{a}_2]$. The term $\mathbf{a}_1^\top W\mathbf{h}_i$ is independent of j, so the relative ordering of neighbours by attention score is the same for every query node — a property that [5] formalises as *static attention*. In other words, if node j_1 is ranked above node j_2 as a neighbour of i, then j_1 is also ranked above j_2 as a neighbour of every other node. For a pose-grouping network this is undesirable: the network should be able to judge that a given left shoulder is important to a right shoulder but not to a left ankle of a different person.

GATv2 resolves the limitation by reordering the operations inside the attention score,

$$e_{ij}^{\text{GATv2}} = \mathbf{a}^\top \,\text{LeakyReLU}(W\left[\mathbf{h}_i \,\|\, \mathbf{h}_j\right]). \tag{3.8}$$

With the nonlinearity now placed inside the query–key interaction, the attention score does not decompose into independent per-node terms, and the ranking of neighbours is allowed to depend on the query node. GATv2 is strictly more expressive than GAT at the same parameter count [5], and every architecture evaluated in this dissertation uses GATv2 as its attention primitive. The remaining

GCN and GAT references in Chapters 4 and 5 refer to baselines and background, not to the working network.

3.2 PRELIMINARIES — CLUSTERING AND LEARNED GROUPING

3.2.1 k-means and constrained k-means

Classical k-means objective and Lloyd's algorithm. Constrained k-means (must-link / cannot-link) as the family that encodes type constraints. Why k-means appears throughout this dissertation as the canonical "simple" clustering head, and the basis for PGA scoring with GT K.

3.2.2 Deep embedded clustering (DEC)

Xie et al. (2016) DEC objective: Student's t-distribution soft assignments q_{ij}, target distribution p_{ij}, KL divergence minimisation. Why DEC was a natural candidate (learn embedding and clustering jointly), and how the alternating refinement works.

3.2.3 Slot attention

Locatello et al. (2020) slot attention: iterative attention between a set of learned slots and input features, soft-binding each input to a slot via competitive softmax. Why slot attention suggested itself for pose grouping (slots as person identities) and what the iteration assumes.

3.2.4 Graph partitioning (MinCut, NormCut)

Graph-cut objectives: minimum cut, normalised cut (Shi & Malik). Relaxation to continuous assignment matrices and the eigenvalue connection. How graph partitioning was applied as a learned grouping head.

3.2.5 Deep modularity networks (DMoN)

Tsitsulin et al. (2023) DMoN objective: Newman modularity as a soft, differentiable community-detection loss. The modularity matrix $B = A - dd^{\top}/2m$ and the collapse-prevention regulariser. Why DMoN was the natural follow-up after the three learned heads failed, and what assumption it makes about the relationship between graph structure and cluster membership (this will matter in Chapter 5).

3.3 PROBLEM FORMULATION

Formalises the task. Input: set of N keypoints with types t_i and 2D positions $\mathbf{p}_i$. Output: a partition into K groups (one per person) with the type constraint. Evaluation metric: Pose Grouping Accuracy (PGA), defined as the Hungarian-matched joint assignment accuracy between predicted and GT groups.

3.4 DATASET PIPELINE

The synthetic data generator (controlled K, perfect GT, procedural skeletons), the COCO keypoints dataset (val2017 used throughout), and the train/val splits. Depth channel handling (retained or removed) and why that decision matters for COCO transfer.

3.5 BASELINE: STANDARD GAT + CONTRASTIVE + KNN

The baseline architecture: stacked GATv2 layers on a fully-connected graph over detected keypoints, contrastive loss pulling same-person keypoints together and pushing different-person keypoints apart, kNN clustering at test time. The depth-input variant and the no-depth variant. This is the starting point of the narrative.

3.6 LEARNED GROUPING HEADS

3.6.1 DEC applied to pose grouping

How DEC was instantiated on top of the GAT embeddings: initialisation strategy, cluster-count handling, training objective combined with contrastive loss.

3.6.2 Slot attention applied to pose grouping

Slot attention head on top of the GAT embeddings, with slots as person identities. Number of slots, initialisation, iteration count, and the binding-to-assignment readout.

3.6.3 Graph partitioning applied to pose grouping

Graph partitioning head on top of the GAT embeddings. Edge-weight construction, normalised-cut objective, and how the soft partition is read out into hard assignments.

3.7 SA-DMON — SKELETON-AWARE DEEP MODULARITY NETWORKS

A DMoN-based learned grouping head adapted for skeletons. v1 (learnable σ, clamped σ, median σ, spectral-term variants) and v2 (feature-only adjacency, entropy-only regularisation, full formulation). Ten experiments in total, all tried to fix the modularity misalignment in a different way.

3.8 PG-GAT — POSE GROUPING GRAPH ATTENTION NETWORK

3.8.1 Design intuition

Reframes the investigation: every learned head failed to beat kNN on the same embeddings, so the bottleneck is the embedding. PG-GAT strengthens the embedding while keeping the grouping head simple (kNN).

3.8.2 Type-pair attention modulation

Per-type-pair learnable bias in the GATv2 attention score, so that different joint-type relationships (e.g. left-shoulder-to-right-shoulder vs left-shoulder-to-head) can be weighted independently.

3.8.3 Skeleton-aware positional encoding

A positional encoding scheme that encodes relative skeleton position (not absolute image position), invariant to translation and scale.

3.8.4 Type-repulsion loss term

An auxiliary contrastive term that explicitly pushes same-type keypoints of different people apart (the difficult case: two left shoulders from different people must not collapse).

3.8.5 Training procedure

Two-stage training: synthetic pretraining (contrastive loss, all three modifications active), then COCO fine-tuning on detected keypoints (20 epochs). Hyperparameters, schedule, optimiser.

3.9 HYPERBOLIC PG-GAT

A variant that embeds the output into a Lorentz-model hyperbolic space. Euclidean internal attention, Lorentz output projection, logmap for clustering. Motivation: skeletons are trees, and trees embed more faithfully in hyperbolic space with fewer dimensions. The variant tests whether this translates into a quality or efficiency advantage for pose grouping.

3.10 TRIGAT — TRIPLET-BASED PRIMITIVE

A variant where the graph nodes are skeleton triplets (A–B–C chains, 19 per person) rather than individual joints. Pivot-angle features are rotation-invariant articulation descriptors. Triplets are embedded, clustered, and joint labels are voted from triplet-cluster assignments. Included to test whether a higher-order primitive improves the embedding.

3.11 INTEGRATION WITH A REALISTIC DETECTOR

How PG-GAT plugs into a HigherHRNet detection pipeline without joint training. Keypoint detection confidence filtering, the K-estimation head (brief treatment — the full version belongs to the PEng follow-up), end-to-end PGA computation.

3.12 EVALUATION PROTOCOL AND METRICS

PGA definition with Hungarian matching. Synthetic evaluation (100-image virtual test set, perfect GT). COCO evaluation zero-shot and after COCO fine-tuning. End-to-end evaluation using HigherHRNet detections. What kNN-PGA measures, what COP-KM-PGA adds, and which numbers are the native MEng headlines (kNN) versus what is mentioned with a pointer to the PEng paper (type-constrained clustering).

CHAPTER 4 EXPERIMENTAL RESULTS

This chapter reports experimental results for every method presented in Chapter 3, in the order of the investigation. Each section follows the lab-book pattern (setup, table of numbers, interpretation). Failed approaches (learned heads, SA-DMoN, TriGAT, visual-feature augmentation) are reported with their numbers *and* explanations of why they failed. Honest caveats are flagged where the numbers are close calls.

4.1 BASELINE: STANDARD GAT

Results for the standard GAT baseline with and without the depth channel. The key observation: removing the MiDaS depth channel *improved* COCO transfer by roughly four points (MiDaS has a synthetic-to-real domain gap). Numbers: 0.994 synth kNN / 0.838 COCO kNN with depth; 0.901 / 0.879 without.

4.2 FAILED LEARNED GROUPING HEADS

4.2.1 DEC

DEC plateaued at 0.654 synth and did not transfer meaningfully to COCO. The Student's-t target distribution concentrates cluster mass but does not separate identities in the pose setting.

4.2.2 Slot attention

Slot attention underperformed kNN on the same embeddings (0.858 synth / 0.788 COCO with the depth baseline). The competitive softmax does not encode the type constraint.

4.2.3 Graph partitioning

Graph partitioning was closest to beating kNN on synthetic (0.954) but collapsed on COCO (0.752). Highly sensitive to edge-weight construction, does not transfer.

4.2.4 Cross-cutting finding

Every learned head failed to beat plain kNN on the same embeddings. This was the first evidence that the lever is the embedding, not the grouping head.

4.3 SA-DMON — TEN EXPERIMENTS

Full results table for all ten SA-DMoN variants (v1 with learnable / clamped / median σ, v1 with λ sweeps, v1 without spectral term, v2 with feature-adjacency variants, v2 with entropy-type-only regularisation, v2 full). Every variant plateaus in the 0.77–0.81 synthetic range. Root cause analysis is deferred to Chapter 5, but the empirical picture is: no SA-DMoN formulation closes the gap to kNN.

4.4 PG-GAT ABLATIONS

Four-row ablation: type-pair-only, pos-enc-only, repulsion-only, full. Each modification contributes independently; the full variant is the sum. Headline synthetic number: 0.910 synth kNN / 0.882 COCO kNN on the synthetic-only training regime. Table with explicit Δ against the standard-GAT-no-depth baseline.

4.5 COCO FINE-TUNING

PG-GAT fine-tuned on COCO keypoints (20 epochs). COCO kNN PGA rises from 0.882 to 0.957. Synthetic PGA rises from 0.910 to 0.928 (the fine-tune is not destructive to synthetic performance). This is the single largest gain of any experiment in the dissertation.

4.6 HYPERBOLIC PG-GAT

Two-part result. Pre-fine-tune: hyperbolic at $d = 32$ matches or slightly beats Euclidean at $d = 128$ (0.918 synth / 0.899 COCO kNN, vs Euclidean 0.910 / 0.882) — consistent with trees needing fewer dimensions in hyperbolic space. Post-fine-tune: the advantage disappears. Hyperbolic $d = 32$ COCO-fine-tuned reaches 0.943 kNN, still below Euclidean $d = 128$ COCO-fine-tuned (0.957). Reported honestly as an *efficiency trade* (comparable quality at $4\times$ smaller embedding), not a quality win.

4.7 TRIGAT — NEGATIVE RESULT

Triplet-based primitive underperforms joint-level PG-GAT on every comparison: 0.898 synth / 0.855 COCO with kNN, versus 0.910 / 0.882 for joint-level. Root cause: triplet clusters must be voted back to joint labels, and small triplet-level errors propagate to multiple joint votes. Documented as a negative result.

4.8 END-TO-END INTEGRATION WITH HIGHERHRNET

Replace the associative-embedding tag head in HigherHRNet with PG-GAT. End-to-end PGA using HigherHRNet-W32-512 for detection: PG-GAT + kNN = 0.942, PG-GAT + COP-Kmeans = 0.955 (COP-KM detail deferred to the PEng paper), HigherHRNet native AE grouping = 0.985. The gap to 0.985 comes from the tight coupling of the tag head to its backbone features, not from embedding-quality issues.

4.9 MODEL SIZE AND MODULARITY

Parameter counts: PG-GAT 170,560; K-head 18,625; HigherHRNet-W32 full model 28,645,331; HigherHRNet AE tag head alone $\sim$561. Correction to the earlier "168$\times$ smaller" claim: PG-GAT is *larger* than the pure AE tag head (170K vs 561). The contribution is modularity — PG-GAT plugs into any detector's keypoint output without joint training; the AE head cannot.

4.10 VISUAL-FEATURE AUGMENTATION (PG-GAT V2) — NEGATIVE RESULT

Added MobileNetV2 features sampled at each keypoint location as additional node input. Result: 0.968 COCO kNN / 0.964 E2E — a 0.8-point E2E improvement at the cost of 2.2M extra parameters. Total pipeline (30.8M) is larger than HigherHRNet (28.6M) while still scoring lower (0.964 vs 0.985). Reported as a negative result; the Paper 3 idea was dropped on this basis.

CHAPTER 5 DISCUSSION

This chapter interprets the results of Chapter 4, answers the research questions from Chapter 1, and examines the limitations of the work. The central question of the chapter is *why* the investigated methods succeeded or failed, not merely what the numbers were.

5.1 WHY THE LEARNED GROUPING HEADS FAILED

Discusses the shared failure mode of DEC, slot attention, and graph partitioning: each added a learned assignment mechanism on top of embeddings that were already close to separable under simple kNN. When the embedding is the bottleneck, stacking a learned head reshapes the assignment space but does not improve separation, and can actively hurt transfer (the learned head overfits to synthetic structure).

5.2 WHY SA-DMON FAILED

Root-cause analysis of the ten SA-DMoN experiments. The modularity objective is fundamentally misaligned with pose grouping: the kNN graph encodes *spatial* proximity, but the target communities are defined by *identity*. The spatial null model either cancels with the spatial adjacency or is miscalibrated, and no null-model reformulation (learnable / clamped / median σ, feature-only adjacency, entropy regularisation) repaired the misalignment. This is a structural, not a tuning, problem.

5.3 WHY PG-GAT WORKS

Discusses what each of the three modifications buys, drawing on the ablation numbers from Chapter 4. Type-pair attention lets the network weight joint-type relationships independently. The skeleton-aware positional encoding provides translation/scale invariance that the contrastive loss alone does not enforce. The type-repulsion term addresses the hardest pairs (same-type, different-person keypoints that look locally similar). Together they make the embedding separable enough that kNN works.

5.4 INTERPRETING THE HYPERBOLIC RESULT

Interprets the dim-32 hyperbolic advantage and its disappearance after fine-tuning. Pre-fine-tune, hyperbolic geometry helps because the model is relying on the geometric prior (trees embed naturally in hyperbolic space). After fine-tuning, the Euclidean model learns a sufficient representation from data, and the geometric prior becomes a constraint rather than a help. The result is a quality-matched efficiency trade, not a quality win.

5.5 WHY TRIGAT UNDERPERFORMED

The triplet primitive is semantically richer per-token but introduces a voting step that propagates errors from clusters to joint labels. A single mis-voted triplet cascades across the three joint votes that triplet contributes to, and the cascaded effect dominates the gap between TriGAT and joint-level PG-GAT in the results of Chapter 4.

5.6 COMPARISON WITH HIGHERHRNET AE — THE MODULARITY TRADE-OFF

Discusses the 0.955 vs 0.985 gap. HigherHRNet's AE head is tiny ($\sim$561 parameters) but inseparable from its backbone; it gets quality from the backbone features being trained jointly with the tag. PG-GAT is larger (170K parameters) but detector-agnostic — a capability the AE head cannot offer. The correct comparison is not "who wins on COCO" but "what does each buy." This framing matters for the modularity contribution claim.

5.7 ANSWERS TO THE RESEARCH QUESTIONS

Explicit answers to RQ1, RQ2, RQ3 from Chapter 1, grounded in the numbers from Chapter 4.

5.8 LIMITATIONS AND THREATS TO VALIDITY

Known limitations: reliance on detector-supplied K (K-estimation is outside the MEng core); synthetic-to-real domain gap mitigated but not eliminated; only one backbone evaluated end-to-end; PGA is Hungarian-matched joint accuracy, not mAP, so comparison to COCO-standard HPE numbers is indirect; COCO val2017 is the only real-image benchmark reported.

CHAPTER 6 CONCLUSION

This chapter summarises the contributions of the dissertation, consolidates the answers to the research questions, and indicates directions for future work. The PEng follow-up is the main pointer forward, but several MEng-local directions are also noted.

6.1 SUMMARY OF CONTRIBUTIONS

Restates the five numbered contributions from Section 1.6, each now grounded in the numbers from Chapter 4. PG-GAT is the main technical contribution. The empirical finding that the embedding dominates the grouping head is the main scientific contribution. The COCO fine-tuning protocol, the hyperbolic variant, and the TriGAT negative result are supporting contributions.

6.2 ANSWERS TO THE RESEARCH QUESTIONS

Condensed one-paragraph answers to RQ1 (yes — PG-GAT + kNN reaches 0.957 COCO), RQ2 (all three PG-GAT modifications contribute independently; COCO fine-tuning adds the largest single improvement), and RQ3 (PG-GAT is 0.03 PGA below HigherHRNet AE end-to-end, but is detector-agnostic — the trade-off is modularity for a small quality cost).

6.3 FUTURE WORK

Three directions. (1) The PEng follow-up re-frames grouping as structured assignment and develops SCOT, SCOT-KI, and THA — closing on HigherHRNet's 0.985 with algorithmic rather than embedding changes. (2) A fully hyperbolic PG-GAT architecture (hyperbolic attention, not only hyperbolic output) would test the geometric-prior hypothesis more cleanly than the partial variant evaluated here. (3) End-to-end joint training with a detector would give up modularity to regain the coupling advantage AE exploits, and would quantify how much of the 0.03 gap is fundamental versus avoidable.

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APPENDIX A DERIVATION OF HIGHER ORDER MODELS

A.1 EXAMPLE HEADING

A.1.1 Subheading example

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